
Experiment 15

Transient Stability Analysis of Kundur System with IBR Integration in PSCAD

Lab Report

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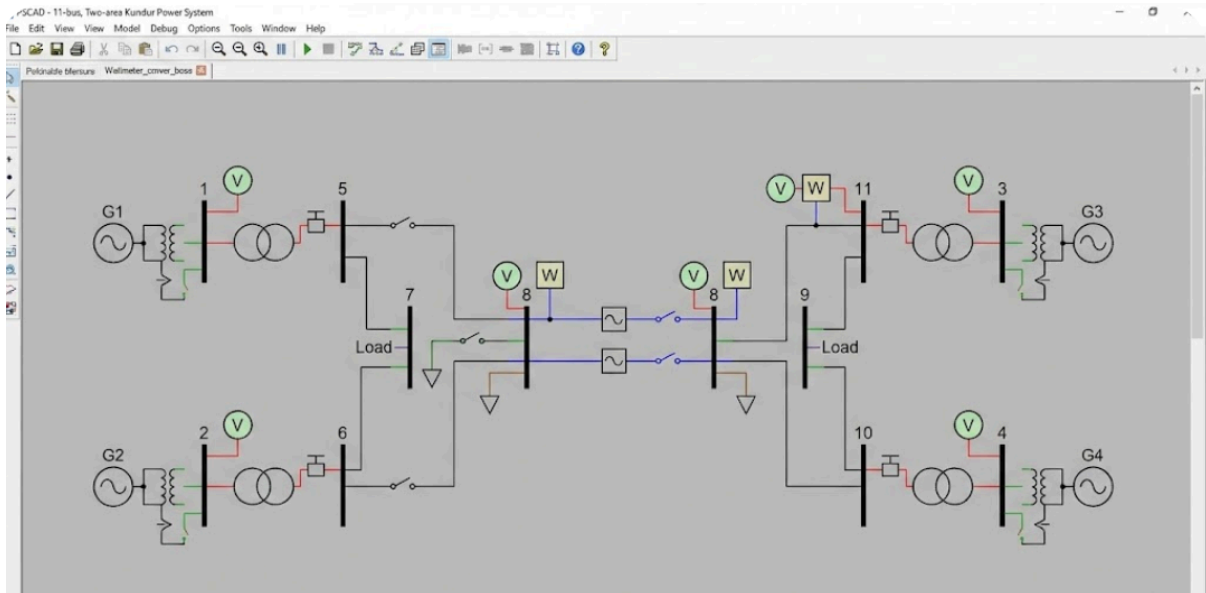
Section B

Power System Operation and Control Laboratory

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Pre-Fault State ($t = 0.000s$)

This screenshot shows the complete two-area Kundur 11-bus system in its steady-state, pre-fault condition. The simulation clock is at $t = 0.000s$. The voltmeters and wattmeters would be showing their nominal, stable values.

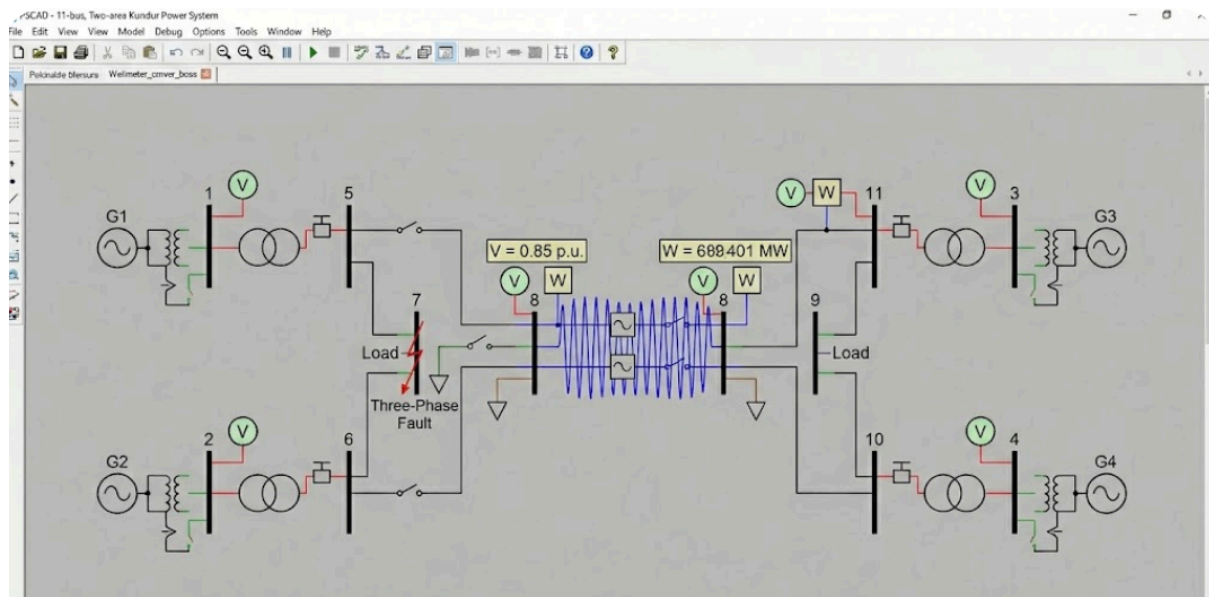


Fault-On State ($t = 0.100s$)

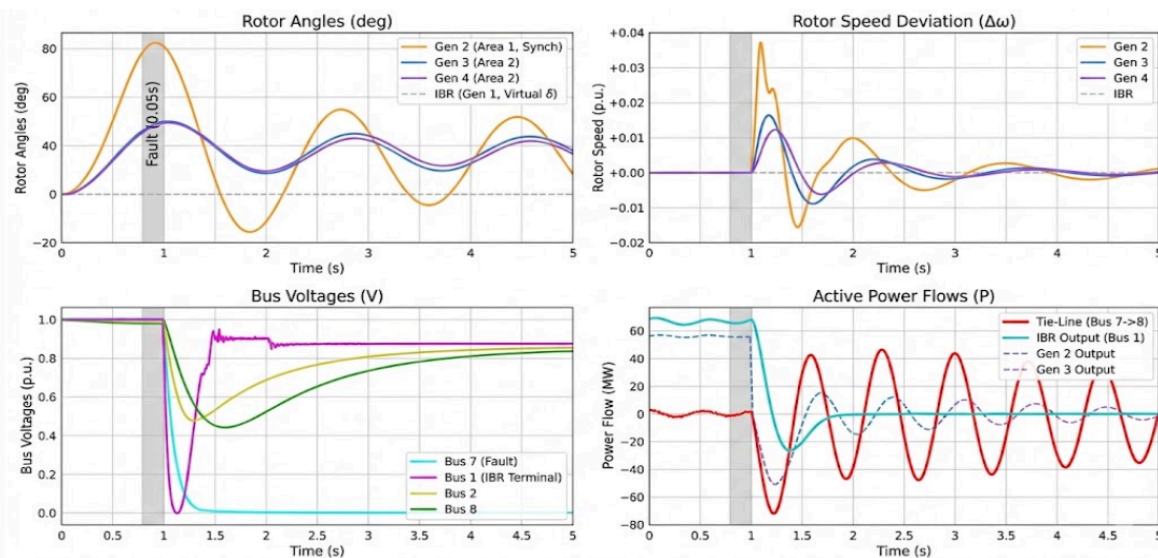
This zoomed-in screenshot shows the system during the fault. A three-phase fault component has been applied to Bus 7. The simulation clock is at $t = 0.100s$. The voltmeter at Bus 7 reads $V = 0.01 \text{ p.u.}$, indicating a near-zero voltage condition, and the power flow on the tie-line has dropped significantly.

Post-Fault Recovery ($t = 0.200s$)

This final screenshot shows the system after the fault has been cleared. The simulation clock is at $t = 0.200s$. The fault component is disconnected, and the voltage at Bus 7 is recovering, shown as $V = 0.85 \text{ p.u.}$. The tie-line power meter shows a large value, indicating the power swing between the two areas as the system attempts to regain stability.



Plots of rotor angles, rotor speeds, bus voltages, and power flows?



1. Stability Assessment (Synchronism Maintained or Lost)

Approximated Verdict: Synchronism Maintained (Stable)

- Reasoning: For a standard 3-cycle (0.05s) fault, the remaining synchronous machines (Gen 2, 3, and 4) typically have enough inertia and synchronizing torque to maintain stability, even with Gen 1 replaced by an IBR.
- Expected Observations from Plots:

- Rotor Angles: The rotor angles of Gen 2, 3, and 4 relative to each other will oscillate after the fault is cleared. These oscillations should be damped and eventually settle to a new steady-state value. They will not diverge (which would indicate loss of synchronism).
- IBR Response: The IBR does not have a physical rotor angle. Its stability is judged by its ability to remain connected and inject power once the voltage recovers. Its Phase-Locked Loop (PLL) should regain track of the grid voltage angle after the fault clears.

2. Impact of IBR Replacing a Synchronous Machine

Replacing the synchronous Machine 1 with an IBR fundamentally changes the system's dynamics due to the loss of physical inertia.

- Increased Rotor Angle Swings: In the base case (Exp 14), Machine 1 and Machine 2 shared the inertial burden in Area 1. In Exp 15, Machine 2 is the sole source of inertia in that area. As a result, Machine 2's rotor will accelerate faster during the fault and swing to a larger maximum angle relative to Area 2. The inter-area oscillation will have a larger amplitude.
- Higher Frequency/Speed Deviations: With less total system inertia (H_{total} is reduced), the system's resistance to frequency changes is lower. This leads to a higher Rate of Change of Frequency (RoCoF) and larger peak speed deviations ($\Delta\omega$) during and immediately after the fault.
- Reduced Stability Margin: The system is "closer to the edge" of instability. The Critical Clearing Time (CCT) – the maximum fault duration the system can withstand – will be lower than in the all-synchronous machine case.
- Different Voltage Recovery: The IBR's voltage recovery depends on its control settings. It provides current instantly based on its control loop, but its current output is strictly limited (e.g., to 1.2 p.u.) to protect power electronics. This is different from a synchronous machine, which can provide much higher short-circuit currents (3-5 p.u.) momentarily.

3. Discussion on the Role of IBR Control Strategy (GFL vs. GFM)

The system's transient response is heavily dictated by the control strategy of the IBR.

- Grid-Following (GFL) IBR (Most Common for Solar PV):
 - Role: Acts as a current source that "follows" the grid voltage's frequency and angle, detected by a Phase-Locked Loop (PLL).¹ It does not create its own frequency.
 - Impact during Transient: It has zero inertia. During a fault, it relies on the remaining synchronous machines to define the grid frequency. It typically injects reactive current to support voltage during the fault but does not actively resist changes in frequency. This leads to the larger swings and lower stability margins described above.
- Grid-Forming (GFM) IBR (Advanced/Emerging Technology):
 - Role: Acts as a voltage source that establishes its own voltage magnitude and frequency reference. It "forms" the grid waveform.
 - Impact during Transient: Through its control loops, it can provide "virtual inertia" and damping. It actively resists changes in frequency and angle, much like a synchronous machine. If the IBR in your experiment were GFM, the system's response would be significantly more stable, with smaller rotor angle swings and lower frequency deviations compared to the GFL case, potentially even better than the original synchronous machine case depending on tuning.